Predicting Avocado Spoilage Risk and Loss Drivers in California

Priyanka Adhikari, Ruchira Banerjee, Nidhi Bendre

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Introduction

This analysis aims to predict monthly avocado spoilage risk across California counties using historical crop insurance data from 2001–2018. Key variables include loss rate, defined as total indemnity paid divided by determined acres in a given month, along with climate related damage causes. Understanding these metrics can help insurers, agricultural planners, and policymakers anticipate periods of elevated crop loss before they occur.

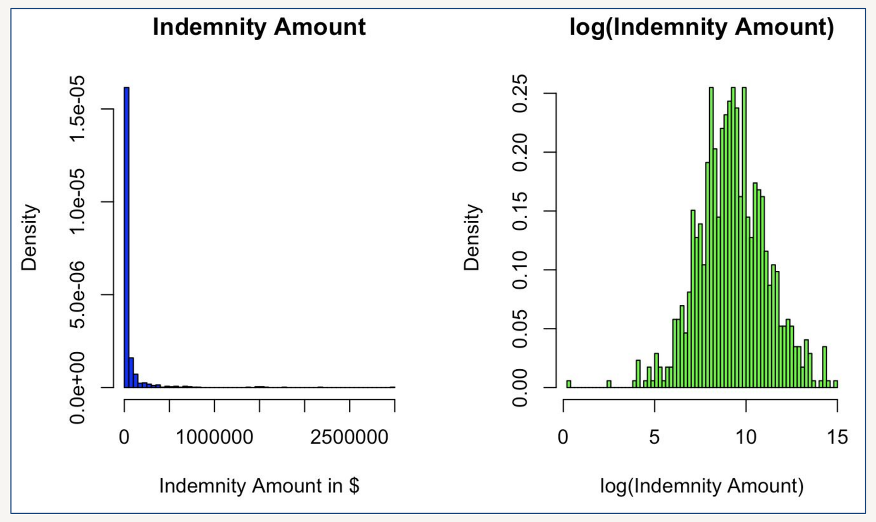
Three ML models are used to address this issue in different angles. The bayesian regression model predicts the insurance claim amounts themselves, the ARIMA model forecasts temporal variability in claims, and the MLP model classifies climate driven damage cause behind each loss.

Data

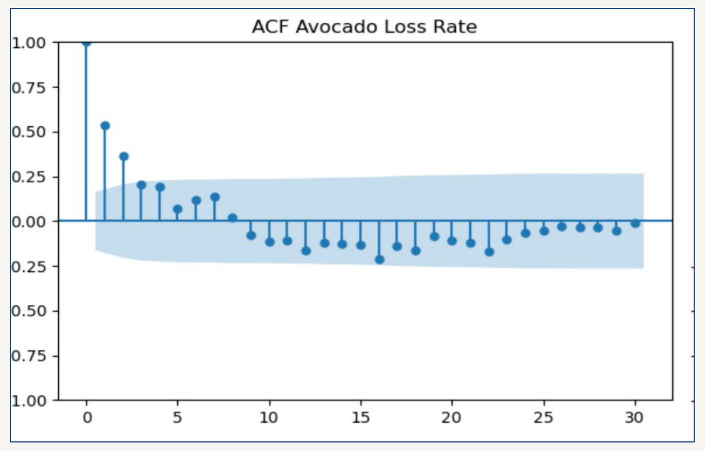
The data used in this analysis comes from joining two sources. Crop loss information comes from the USDA Risk Management Agency historical Cause of Loss archives spanning 2001-2018, which details indemnities paid to producers for crop failures. Data covers extreme weather events such as drought, excess moisture, hail, wind, and frost. Records were then filtered to focus on avocado loss in California. Climate data was obtained from NOAA National Centers for Environmental information statewide climate at a glance portal which provides monthly statewide climate summaries for average temperature, total precipitation, heating/cooling degree days, and the Palmer Drought Severity Index (PDSI). The two sources were joined on month/year to produce a single dataset of 871 records.

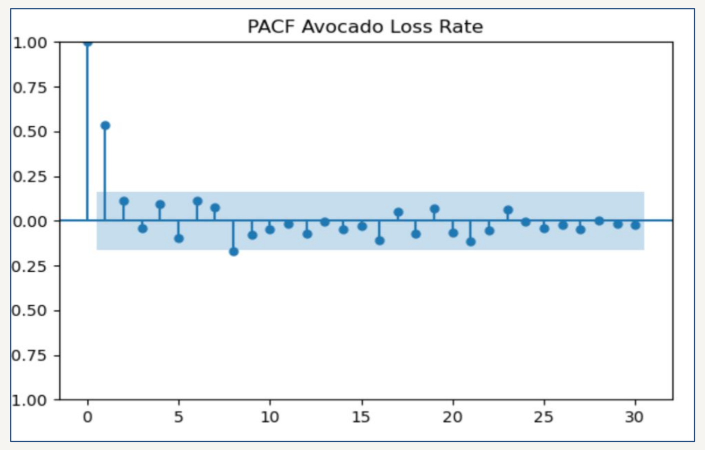
Methods

Bayesian: To predict log transformed avocado insurance claim amounts, this model was fit using brms with a Gaussian likelihood, weakly informative priors, and random intercepts for categorical features. Posterior sampling was performed with 4 chains and 2000 iterations.



ARIMA: To forecast monthly spoilage rate based on the claims data, time series approach was employed using an ARIMA model. The ACF and PACF plots for avocado loss rate both showed a strong spike at lag 1 followed by gradual decay across higher lags, indicating that an ARIMA model would be best.

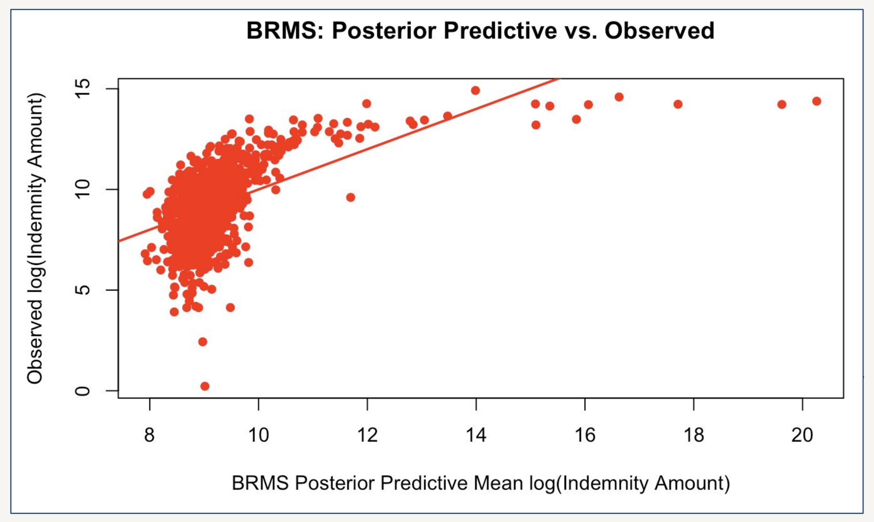


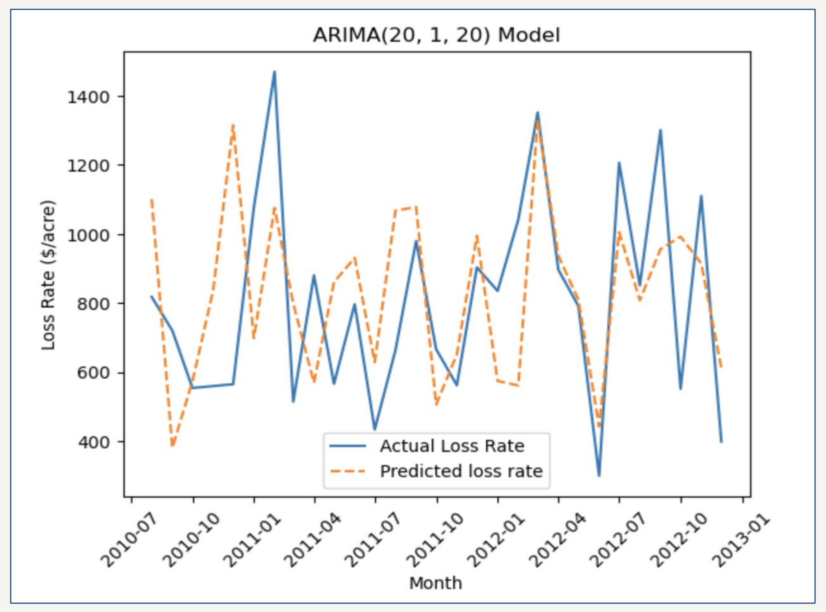


MLP: To classify damage cause behind loss, a two layer MLP model with ReLU activations in both hidden layers and a softmax output trained with cross entropy loss was implemented. Features included climate variables, loss month, county, and insurance plan.

Results

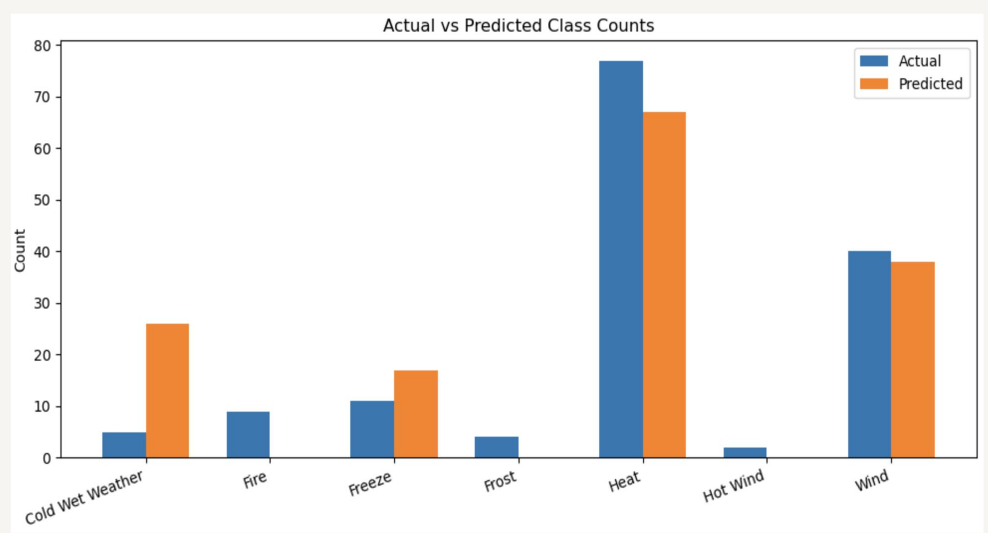
Bayesian: R^2 = 0.31 with an RMSE of 1.55, with determined acres as the primary feature of interest. Temperature trended negative and PDSI (drought index) indicated that drier conditions come with larger





ARIMA: Based on multiple iterations, ARIMA(20,1,20) was selected as the final model, achieving a test MAE of 237.99 and RMSE of 289.27. However, its complexity risks overfitting, and adding weather variables could improve generalization.

MLP: Achieved around 50.68% accuracy and the plot converged steadily over 2000 epochs. Per-class results reveal a pattern tied to climate distinctiveness. Heat (66.2%), freeze (63.6%), and cold wet weather (60.0%) were predicted most accurately. Fire, frost, and hot wind registered 0%.



Conclusion

Overall, the models show that avocado spoilage risk is partially predictable, though performance varies. The Bayesian model highlights determined acres as a key driver and suggests drier conditions increase losses, with some uncertainty. ARIMA captures the volatility of loss rates well but risks overfitting due to its complexity. The MLP achieves moderate accuracy, performing best on distinct climate events like heat and freeze, but struggles with less distinct or rare categories.

Future Steps

To improve upon these results, a meaningful next step would be to integrate the models by using the MLP's predicted damage cause as an additional feature in the Bayesian regression, since damage type was shown to explain meaningful group-level variation. Along with this, incorporating lagged weather variables in a SARIMAX model would hep improve the spike-prediction limitation identified in the time series model. Finally, expanding beyond avocados to other produce crops would test whether these patterns generalize or are specific to avocado growing conditions in California.

References

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